

PURPLLE ONLINE BEAUTY SHOPPING

SELLER AND INVENTORY MANAGEMENT

1. Introduction

The **Purple Online Beauty Shopping – Seller and Inventory Management System** is designed to manage seller information, products, and stock details efficiently. The system uses related database tables such as **Seller, Product, and Inventory** with primary and foreign keys to maintain data integrity. It supports maintaining seller-product information, tracking available and unavailable products, updating stock quantities, and generating inventory reports.

2. Objectives

- To create and maintain **seller information**.
- To maintain **seller-product relationships**.
- To record and manage **product stock quantities**.
- To track **available and unavailable products**.
- To update inventory details whenever stock changes.
- To generate **inventory status reports**.
- To maintain data integrity using **Primary Key and Foreign Key constraints**.
- To generate **seller-wise total stock reports**.

3. Database Design

3.1 Seller Table

Attribute	Data Type	Key	Description
Seller_ID	NUMBER	Primary Key	Unique seller identifier
Seller_Name	VARCHAR2(100)	-	Name of the seller (not null)
Email	VARCHAR2(100)	-	Seller email (unique)
Phone	VARCHAR2(15)	-	Seller contact number
GSTIN	VARCHAR2(20)	-	GST identification number (unique)
Warehouse_Address	VARCHAR2(255)	-	Warehouse location
Rating	NUMBER(3,2)	-	Seller rating

3.2 Inventory Table

Attribute	Data Type	Key	Description
Inventory_ID	NUMBER	Primary Key	Unique inventory record identifier
Product_ID	NUMBER	Foreign Key	References PurpllerProduct(Product_ID)
Seller_ID	NUMBER	Foreign Key	References Seller(Seller_ID)
Stock_Quantity	NUMBER	-	Quantity in stock (≥ 0)
Product_Status	VARCHAR2(20)	-	'Available' or 'Unavailable'
Last_Updated	DATE	-	Date the stock was last updated

4. Table Relationships

Seller and Inventory have a one-to-many (1:M) relationship — one seller supplies many inventory records. PurpllerProduct and Inventory also have a one-to-many (1:M) relationship — one product can have inventory records from different sellers.

SELLER (1) ————— (M) INVENTORY (M) ————— (1)
PURPLLEPRODUCT
Seller_ID (PK) Seller_ID (FK) / Product_ID (FK) Product_ID (PK)

5. Table Creation

5.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Email VARCHAR2(100) UNIQUE,  
5     Phone VARCHAR2(15),  
6     GSTIN VARCHAR2(20) UNIQUE,  
7     Warehouse_Address VARCHAR2(255),  
8     Rating NUMBER(3,2)  
9 );  
  
Table created.
```

5.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER NOT NULL,  
4     Seller_ID NUMBER NOT NULL,  
5     Stock_Quantity NUMBER NOT NULL CHECK (Stock_Quantity >= 0),  
6     Product_Status VARCHAR2(20),  
7     Last_Updated DATE,  
8     FOREIGN KEY (Product_ID) REFERENCES PurpleProduct(Product_ID),  
9     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
10 );  
  
Table created.
```

6. Seller Data Insertion

```
SQL> INSERT INTO Seller VALUES (301, 'Lakme Seller', 'lakme@gmail.com', '9876543210', 'GSTIN301', 'Chennai', 4.50);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (302, 'Maybelline Seller', 'maybelline@gmail.com', '9876543211', 'GSTIN302', 'Bangalore', 4.40);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (303, 'Mamaearth Seller', 'mamaearth@gmail.com', '9876543212', 'GSTIN303', 'Chennai', 4.60);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (304, 'Minimalist Seller', 'minimalist@gmail.com', '9876543213', 'GSTIN304', 'Mumbai', 4.70);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (305, 'Loreal Seller', 'loreal@gmail.com', '9876543214', 'GSTIN305', 'Delhi', 4.30);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (306, 'Tresemme Seller', 'tresemme@gmail.com', '9876543215', 'GSTIN306', 'Chennai', 4.20);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (307, 'Fogg Seller', 'fogg@gmail.com', '9876543216', 'GSTIN307', 'Bangalore', 4.30);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (308, 'Bella Vita Seller', 'bellavita@gmail.com', '9876543217', 'GSTIN308', 'Mumbai', 4.20);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (309, 'Dove Seller', 'dove@gmail.com', '9876543218', 'GSTIN309', 'Delhi', 4.50);  
1 row created.  
  
SQL> INSERT INTO Seller VALUES (310, 'Nivea Seller', 'nivea@gmail.com', '9876543219', 'GSTIN310', 'Chennai', 4.40);  
1 row created.  
  
SQL> COMMIT;  
Commit complete.
```

7. Inventory Data Insertion

```
SQL> INSERT INTO Inventory VALUES (1, 101, 301, 50, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (2, 102, 302, 75, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (3, 103, 303, 60, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (4, 104, 304, 40, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (5, 105, 305, 80, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (6, 106, 306, 55, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (7, 107, 307, 35, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (8, 108, 308, 30, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (9, 109, 309, 90, 'Available', SYSDATE);  
1 row created.  
  
SQL> INSERT INTO Inventory VALUES (10, 110, 310, 65, 'Available', SYSDATE);  
1 row created.  
  
SQL> COMMIT;  
Commit complete.
```

8. Data Retrieval

8.1 Display Seller Records

```
SQL> SELECT * FROM Seller;
```

```
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
301  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
Lakme Seller  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
lakme@gmail.com  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
9876543210      GSTIN301
```

```
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
Chennai  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
4.5  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----  
302  
  
SELLER_ID  
-----  
SELLER_NAME  
-----  
EMAIL  
-----  
PHONE          GSTIN  
-----  
WAREHOUSE_ADDRESS  
-----  
RATING  
-----
```

```
WAREHOUSE_ADDRESS
-----
RATING
-----
minimalist@gmail.com
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543213 GSTIN304
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
Mumbai
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
4.7
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
SELLER_ID
```

```
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543212 GSTIN303
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
Chennai
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
4.6
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
-----
RATING
-----
304
SELLER_ID
```

```
-----
RATING
-----
4.4
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
303
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Mamaearth Seller
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
mamaearth@gmail.com
SELLER_ID
-----
SELLER_NAME
-----
302
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Maybelline Seller
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
maybelline@gmail.com
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543211 GSTIN302
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Bangalore
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
```

```

SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
nivea@gmail.com
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
9876543219 GSTIN318
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
Chennai
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
4.4
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
18 rows selected.

```

```

PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
389
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
Dove Seller
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
dove@gmail.com
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING
9876543218 GSTIN389
SELLER_ID
SELLER_NAME
EMAIL
PHONE GSTIN
WAREHOUSE_ADDRESS
RATING

```

```
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
308

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Bella Vita Seller

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
bellavita@gmail.com

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543217      GSTIN308

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
```

```
SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
307

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Fogg Seller

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
fogg@gmail.com

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543216      GSTIN307

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
```

```

RATING
-----
306

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
resemme Seller

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
resemme@gmail.com

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
876543215      GSTIN306

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Chennai

SELLER_ID
-----
SELLER_NAME
-----

```

```

WAREHOUSE_ADDRESS
-----
RATING
-----
305

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Loreal Seller

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
loreal@gmail.com

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
9876543214      GSTIN305

SELLER_ID
-----
SELLER_NAME
-----
EMAIL
-----
PHONE          GSTIN
-----
WAREHOUSE_ADDRESS
-----
RATING
-----
Delhi

```

8.2 Display Inventory Records

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	PRODUCT_STATUS	LAST_UPDA
1	101	301	50	Available	17-SEP-26
2	102	302	75	Available	17-SEP-26
3	103	303	60	Available	17-SEP-26
4	104	304	40	Available	17-SEP-26
5	105	305	80	Available	17-SEP-26
6	106	306	55	Available	17-SEP-26
7	107	307	35	Available	17-SEP-26
8	108	308	30	Available	17-SEP-26
9	109	309	90	Available	17-SEP-26
10	110	310	65	Available	17-SEP-26

```
10 rows selected.
```

9. Inventory Update

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 25,
  3     Product_Status = 'Available',
  4     Last_Updated = SYSDATE
  5 WHERE Inventory_ID = 1;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.

SQL> UPDATE Inventory
  2 SET Stock_Quantity = Stock_Quantity + 20,
  3     Last_Updated = SYSDATE
  4 WHERE Inventory_ID = 3;

1 row updated.

SQL>
SQL> COMMIT;

Commit complete.
```

```
SQL> SELECT * FROM Inventory;
```

INVENTORY_ID	PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	PRODUCT_STATUS	LAST_UPDA
1	101	301	25	Available	17-SEP-26
2	102	302	75	Available	17-SEP-26
3	103	303	80	Available	17-SEP-26
4	104	304	40	Available	17-SEP-26
5	105	305	80	Available	17-SEP-26
6	106	306	55	Available	17-SEP-26
7	107	307	35	Available	17-SEP-26
8	108	308	30	Available	17-SEP-26
9	109	309	90	Available	17-SEP-26
10	110	310	65	Available	17-SEP-26

```
10 rows selected.
```

10. Seller Product Information

10.1 Display Seller and Products Details

```
SQL> SELECT
2      S.Seller_ID,
3      S.Seller_Name,
4      P.Product_ID,
5      P.Product_Name,
6      I.Stock_Quantity,
7      I.Product_Status
8  FROM Seller S
9  JOIN Inventory I
10 ON S.Seller_ID = I.Seller_ID
11 JOIN PurpileProduct P
12 ON I.Product_ID = P.Product_ID
13 ORDER BY S.Seller_ID;

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
301
Lakme Seller
101

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Lakme Foundation
25 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
302
Maybelline Seller
101

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Maybelline Lipstick
75 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
303
Mamearth Seller
103

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Mamearth Face Mask
80 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
304
Minimalist Seller
104

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Minimalist Serum
40 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
305
Loreal Seller
105

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS
-----
Loreal Shampoo
80 Available

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
```

```

=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
386
Fresemme Seller
106
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Fresemme Conditioner
55 Available
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
387
Fogg Seller
107
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Fogg Perfume
35 Available
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
388
Bella Vita Seller
108
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Nivea Body Lotion
30 Available
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====

```

```

SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Nivea Body Lotion
30 Available
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
389
Dove Seller
109
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Dove Body Wash
90 Available
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
310
Nivea Seller
110
SELLER_ID
=====
SELLER_NAME
=====
PRODUCT_ID
=====
PRODUCT_NAME
=====
STOCK_QUANTITY PRODUCT_STATUS
=====
Lakme Kajal
65 Available
10 rows selected.
SQL>

```

11. Inventory Status

11.1 Display Available Products

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     S.Seller_Name,
  5     I.Stock_Quantity
  6 FROM Inventory I
  7 JOIN PurplleProduct P
  8 ON I.Product_ID = P.Product_ID
  9 JOIN Seller S
10 ON I.Seller_ID = S.Seller_ID
11 WHERE I.Product_Status = 'Available';
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      101
Lakme Foundation
Lakme Seller
      25

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      102
Maybelline Lipstick
Maybelline Seller
      75

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      103
Mamaearth Face Wash
Mamaearth Seller
      80

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
Loreal Shampoo
```

```

STOCK_QUANTITY
-----
      106
Tresemme Conditioner
Tresemme Seller
      55

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      107
Fogg Perfume
Fogg Seller
      35

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      108
Nivea Body Lotion
Bella Vita Seller
      30

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      109
Dove Body Wash
Dove Seller
      90

PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY
-----
      110
Lakme Kajal
Nivea Seller
      65

10 rows selected.

```

11.2 Display Unavailable Products

```
SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     S.Seller_Name,
  5     I.Stock_Quantity
  6 FROM Inventory I
  7 JOIN PurpleProduct P
  8 ON I.Product_ID = P.Product_ID
  9 JOIN Seller S
 10 ON I.Seller_ID = S.Seller_ID
 11 WHERE I.Product_Status = 'Unavailable';

no rows selected
```

12. Inventory Status Reports

12.1 Complete Inventory Report

```
SQL> SELECT
  2     I.Inventory_ID,
  3     S.Seller_Name,
  4     P.Product_Name,
  5     I.Stock_Quantity,
  6     I.Product_Status,
  7     I.Last_Updated
  8 FROM Inventory I
  9 JOIN Seller S
 10 ON I.Seller_ID = S.Seller_ID
 11 JOIN PurpleProduct P
 12 ON I.Product_ID = P.Product_ID
 13 ORDER BY I.Inventory_ID;
```

```
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
1
Lakme Seller
Lakme Foundation
25 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
2
Maybelline Seller
Maybelline Lipstick
75 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
3
Mamaearth Seller
Mamaearth Face Wash
80 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
4
Minimalist Seller
Minimalist Serum
40 Available      17-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
5
```

```

Tresemme Conditioner
      55 Available          17-SEP-26

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
      7
Fogg Seller
Fogg Perfume
      35 Available          17-SEP-26

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
      8
Bella Vita Seller
Nivea Body Lotion
      30 Available          17-SEP-26

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
      9
Dove Seller
Dove Body Wash
      90 Available          17-SEP-26

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY PRODUCT_STATUS      LAST_UPDA
-----
     10
Nivea Seller
Lakme Kajal
      65 Available          17-SEP-26

10 rows selected.

```

12.2 Available and Unavailable Product Count

```

SQL> SELECT
      2      Product_Status,
      3      COUNT(*) AS Product_Count
      4 FROM Inventory
      5 GROUP BY Product_Status;

PRODUCT_STATUS      PRODUCT_COUNT
-----
Available          10

```

12.3 Total Stock by Seller

```
SQL> SELECT
  2     S.Seller_ID,
  3     S.Seller_Name,
  4     SUM(I.Stock_Quantity) AS Total_Stock
  5 FROM Seller S
  6 JOIN Inventory I
  7 ON S.Seller_ID = I.Seller_ID
  8 GROUP BY S.Seller_ID, S.Seller_Name
  9 ORDER BY S.Seller_ID;
```

```
SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      301
Lakme Seller
      25

      302
Maybelline Seller
      75
```

```
SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      303
Mamaearth Seller
      80

      304
Minimalist Seller
```

```
SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
```

```

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      40

      305
Loreal Seller
      80

      306

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
Tresemme Seller
      55

      307
Fogg Seller
      35

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----
      308
Bella Vita Seller
      30

      309
Dove Seller
      90

SELLER_ID
-----
SELLER_NAME
-----
TOTAL_STOCK
-----

      310
Nivea Seller
      65

10 rows selected.

```

12.4 Products with Zero Stock

```

SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     S.Seller_Name,
  5     I.Stock_Quantity,
  6     I.Product_Status
  7 FROM Inventory I
  8 JOIN PurpleProduct P
  9 ON I.Product_ID = P.Product_ID
 10 JOIN Seller S
 11 ON I.Seller_ID = S.Seller_ID
 12 WHERE I.Stock_Quantity = 0;

no rows selected

```

13. Conclusion

The Seller and Inventory Management System successfully manages seller, product, and stock information in the Purple database. Using SQL operations and table relationships, the system can track product availability, update inventory quantities, and generate useful inventory reports. The use of **Primary Keys and Foreign Keys** helps maintain data consistency and reliable relationships between sellers, products, and inventory.